DENIS SHTEPA

Research-Oriented System Designer & Team Lead — AI-Native Architecture (KG / MCP / RAG)

Tomsk, Russia | Email: ischademadda@gmail.com | Telegram: @ischademadda | GitHub: github.com/ischademadda
| BSc Software Engineering (TPU, expected 2027, GPA: 3.93/5.0)

Research-oriented System Designer and Team Lead with 2+ years leading an engineering R&D team (4–7 engineers). Built **archmap** from 0 to 1 at T-Bank (covered about 40 microservice repositories, generating C4 architecture documentation in about 15 minutes per repository); now leading the design and early implementation of **Archon**, an AI-native architecture knowledge platform (diploma project). Focused on AI-native architecture — LLM pipelines with verifiable Knowledge Graphs, MCP-based agent context — and applied systems research. Seeking a Master's in Computer Science / Software Engineering (Fall 2027).

LEADERSHIP & PROFESSIONAL EXPERIENCE

Team v3.9 — Applied R&D Lab **Tomsk, Russia**
Founder, Team Lead & System Designer (Team: 4–7 engineers) *Sep 2024 — Present*

1. Archon — AI-Native Source of Truth & Architecture Platform (Diploma, in development) `[Kotlin, Python, Temporal, PostgreSQL + AGE, Qdrant, LikeC4]`

- **Core Product Vision:** Addressing architectural drift by designing an automated Knowledge Graph platform that synchronizes C4 architecture with code and feeds verified context to AI engineering agents via MCP.
- **System Design:** Decomposed the system into 17 logical modules packaged as 3 polyglot services orchestrated by Temporal Workflows (ADR-0007); authored 7 ADRs on core trade-offs (multi-tenancy, packaging, MCP) prior to implementation. Base skeleton in progress.
- **Code-to-Architecture Pipeline:** Designed a hybrid extraction pipeline: deterministic AST parsing (Tree-sitter) + 2-phase LLM extraction with a strict "Proof-of-Truth" rule (discarding unproven graph edges) and a token-saving File Selector stage.
- **Scalable Graph Storage:** Selected Graph-per-Tenant isolation on PostgreSQL 16 + Apache AGE (ADR); defined NFR for Impact Analysis traversal (<100ms at depth 4), to be validated in benchmarks.

2. OmniSearch Engine — On-Premise Multimodal Video RAG System (R&D) `[Kotlin Ktor, Go, Python Whisper/CLIP, Qdrant, MinIO S3]`

- **System Architecture:** Separated Control Plane (Kotlin Ktor) and Compute Plane (Go worker pool, Python ML nodes); defined OpenAPI 3.0 contracts and asynchronous media processing finite-state machine.
- **Applied Scenario:** Built a working prototype of on-premise semantic video search (Whisper audio transcription + CLIP frame embeddings) for safety compliance monitoring; demonstrated it live to **Gazprom Vostok Neft** specialists (prototype only, not deployed in production).

3. Archonity — Static Execution-Path Analyzer (TechArena 2026 hackathon, Huawei R&D case) `[Rust, Python, Tree-sitter, local LLM via llama.cpp]`

- **Engineering:** Co-developed (team of 3; ~285 of ~590 commits) a static analyzer predicting which methods, calls, remote requests and SQL tables a user scenario touches in the 41-microservice Train Ticket benchmark: deterministic Rust engine, a local LLM that only maps scenario text to endpoints, file-and-line evidence for every predicted edge.

T-Bank **Moscow, Russia (Remote)**
System Analyst (Junior+, Full-Cycle System Design) *Apr 2026 — Aug 2026 (Internship, Full-Cycle System Design)*

- **archmap Platform (Sole Developer & Owner):** After a platform lead voiced the need for a direction map, built an automated C4 architecture generator solo, aligned with leads. Combined AST extraction, LLMs, and LikeC4 with CI/CD integration. Generated C4 documentation in about 15 minutes per repository; approved by management and scaled to about 40 repositories.
- **Cross-System Integrations:** Designed end-to-end specifications for co-branding features across 3 external systems with backward-compatible REST APIs; eliminated delivery blockers at the specification phase.
- **Operational Efficiency:** Refactored on-call runbooks across 145 incidents, reducing mean response time by 40%.

CORE COMPETENCIES & SKILLS

Product & Architecture: System Design, C4 Model, LikeC4, OpenAPI/REST, ADR/RFC, BPMN, Use Cases, NFR, Impact Analysis.

Distributed Systems & AI: Microservices, Event-Driven Architecture, Temporal, Knowledge Graphs (Apache AGE), Vector DBs (Qdrant), RAG, Model Context Protocol (MCP).

Languages & Technologies: Kotlin, Go, Python, TypeScript, PostgreSQL, Redis, Docker, Git | English (IELTS Academic scheduled 5 Dec 2026), Chinese HSK (learning).